

MM-ABC: Towards Generalist Mobile Manipulation via Seeing, Coordinating and Imagining

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September 2026

ABSTRACT

Mobile manipulation extends robot interaction beyond a fixed kinematic workspace by making the reachable region itself controllable. This flexibility introduces two central challenges: spatially grounded perception under continuous ego-motion and coordinated control of heterogeneous arm and base actions. Existing approaches strengthen geometry through explicit 3D representations or predictive world models, and often decouple mobility and manipulation into separate action streams. We argue that effective mobile manipulation requires not only decoupling, but also representations that support efficient cross-stream collaboration. We present **MM-ABC**, a foundation model built around **Seeing, Coordinating, and Imagining Arm-Base Collaboration**. MM-ABC combines sparse multi-level VLM features for spatial perception; a training-only future branch that uses world imagination and geometric intent as extra supervision, strengthening perception and manipulation-intent prediction and improving the overall learning signal; and MM-APT, which coordinates separate manipulation and mobility streams through masked joint attention and clean-action x -prediction. In controlled ablations, replacing clean-action prediction with velocity prediction lowers success on RoboCasa365 composite-seen tasks from 32.8% to 29.2%, and removing future supervision or multilevel conditioning causes larger drops. We pretrain MM-ABC on **5,000+ hours** of heterogeneous robot data spanning **400K+ episodes, 12 datasets, and 17 embodiments**. Experiments cover EBench, RoboCasa365, ManiSkill-HAB, LIBERO, LIBERO-Plus, and real-world mobile manipulation. MM-ABC achieves 44.71% success on EBench, 61.2% on RoboCasa365, 99.1% on LIBERO, 82.8% on LIBERO-Plus without perturbation training, and 83% mean success on five real-world tasks.

KEYWORDS Mobile Manipulation • Spatial Understanding • Arm-Base Collaboration

1 Introduction

Robot manipulation ultimately requires bringing the robot’s joints and end effectors into suitable configurations to interact with objects in the physical world. For a fixed-base manipulator, however, this interaction is fundamentally bounded by its kinematic workspace: once a target lies outside the reachable region, even a capable policy cannot interact with it. A mobile base changes this constraint. By actively repositioning the robot, it turns the reachable workspace from a fixed property into a controllable variable, extending manipulation from local tabletops to rooms and large-scale environments. From this perspective, *mobile manipulation is not merely navigation followed by manipulation; it is manipulation over a dynamically reconfigurable workspace whose extent the policy itself controls.*

This expanded workspace introduces two central challenges. First, **spatial perception becomes dynamic**: ego-motion continuously changes viewpoints and reference frames, while interaction still requires precise relative geometry among the robot, target, and scene. Second, **arm–base actions must be coordinated**. The base performs large-scale repositioning, whereas the manipulator executes fine-grained interaction. Their dynamics and temporal roles differ substantially, yet they are inherently coupled: base motion determines what the arm can reach, while the intended manipulation determines where the base should move.

Existing work addresses the spatial challenge through point clouds, depth, geometric tokens, dynamic-aware 3D representations, spatial memories, or auxiliary geometric objectives [1–4]. Predictive models further learn future visual or geometric representations to improve action learning [5–7]. While effective, explicit reconstruction or dense future generation can introduce substantial modeling and inference cost. We instead ask whether *world imagination* and *geometric intent* can serve purely as extra training supervision: denser geometric targets that improve perception and manipulation-intent prediction, and thereby the overall learning signal, without requiring a world model at deployment.

The coordination problem raises a complementary question. Recent methods increasingly decouple mobility and manipulation through separate action branches, subsystem-specific perception, or structured action spaces [1, 2, 6, 8, 9]. However, *decoupling alone does not guarantee collaboration*. Once arm and base are represented by different streams, the key design question becomes how these streams should exchange information while preserving their own physical structure and distinct control semantics.

This motivates us to revisit the output parameterization of flow-based action generation. Under rectified flow, clean-action x -prediction and velocity v -prediction are algebraically equivalent, but they need not impose the same representational burden on a finite-width multi-stream expert. A velocity head must retain and cancel high-dimensional flow noise to recover the endpoint, whereas a clean-action head can map directly toward the action manifold. In a controlled synthetic study with a known Bayes-optimal denoiser, x -prediction retains far less injected noise in its endpoint estimates and denoises more accurately at high noise. Under few-step sampling, it also realizes the prescribed arm–body task allocation more accurately. These observations motivate clean-action prediction in our two-stream action expert, whose streams share a finite width.

Building on these observations, we introduce **MM-ABC**, organized around three principles: **Seeing, Coordinating, and Imagining**. For **seeing**, MM-ABC uses a sparse *DeepStack* interface that injects multi-level VLM features into the action expert, preserving complementary fine-grained and semantic information. For **imagining**, learnable future queries predict geometry-rich representations at sparse horizons under a frozen geometric teacher, capturing workspace evolution and the geometric intent the policy should realize. This supervision densifies training signals for current-scene perception and manipulation-intent prediction, while future representations never access action tokens and the branch is removed at deployment. For **coordinating**, we introduce the **Mobile Manipulation Action Prediction Transformer (MM-APT)**, which maintains separate manipulation and mobile/body streams while enabling cross-stream interaction through masked joint attention. MM-APT further combines clean-action x -prediction with asymmetric *near–far* attention: executable near-term actions cannot attend to speculative far-future actions, while later actions build upon earlier ones.

To scale MM-ABC across embodiments, we construct a heterogeneous pretraining mixture containing **5,000+**

hours of robot data, spanning **12 datasets**, **51 subsets**, **400K+ episodes**, and **17 embodiments**. We further collect **MM-30**, a real-world mobile manipulation dataset with **30+ hours** of multi-view demonstrations on a HexFellow Trigger-A3 omnidirectional base with two AgileX PiPER-X 6-DoF arms, covering **40+ tasks** with broad object, background, and skill diversity. A unified canonical action space and embodiment-aware masking preserve embodiment-specific control structure across heterogeneous sources.

We evaluate MM-ABC on **EBench**, **RoboCasa365**, **ManiSkill-HAB**, **LIBERO**, and **LIBERO-Plus**, together with five real-world mobile manipulation tasks (Section 6). MM-ABC achieves 44.71% success on EBench and a task-weighted average of 61.2% on RoboCasa365, exceeding the strongest baselines in the respective comparisons by 3.30 and 7.0 percentage points. It also achieves 99.1% mean success on LIBERO, and the same LIBERO-trained policy reaches 82.8% on LIBERO-Plus without further training, the highest among the compared methods, including those trained on perturbed demonstrations. On five real-world tasks, MM-ABC achieves 83% mean success, 12 points above $\pi_{0.5}$, supporting the use of a shared architecture across fixed-base and mobile settings. These results suggest that scalable mobile manipulation requires more than adding mobility to a manipulation policy: the model must *see* a changing workspace, *imagine* how interaction will reshape it, and *coordinate* arm and base through representations designed for collaboration.

Our contributions are summarized as follows:

- We present **MM-ABC**, a large-scale foundation model for mobile manipulation built around Seeing, Coordinating, and Imagining Arm–Base Collaboration.
- We introduce sparse **DeepStack** visual conditioning and training-only **world imagination**, using future geometry as extra supervision of geometric intent so that perception and manipulation-intent prediction receive a denser learning signal without deployment-time overhead.
- We propose **MM-APT**, a structured dual-stream action transformer with clean-action x -prediction and near–far attention. A controlled synthetic study and a component ablation on RoboCasa365 both favor clean-action prediction over velocity prediction.
- We collect **MM-30**, a real-world mobile manipulation dataset covering **40+ tasks** and **30+ hours** of multi-view demonstrations with diverse objects, backgrounds, and everyday skills.
- We pretrain on **5,000+ hours**, **400K+ episodes**, **12 datasets**, and **17 embodiments**, and evaluate mobile and fixed-base manipulation through simulation benchmarks, controlled ablations, and real-world deployment on five household, office, workcell, and laboratory tasks. MM-ABC delivers strong results throughout, with clear leads on the mobile manipulation benchmarks, strong fixed-base performance on LIBERO and LIBERO-Plus, and 83% mean success in the real world.

2 Related Work

2.1 Mobile Manipulation

Mobile manipulation couples base motion with object interaction, extending robot control beyond a fixed workspace. Early model-based whole-body controllers [10] were followed by reinforcement and imitation learning approaches [9, 11–14], including whole-body coordination for dynamic grasping with legged manipulators [15]. Generalist policies now support discrete action decoding [16, 17], diffusion [18], and flow matching [19], with mobile extensions addressing unfamiliar environments [20], trajectory optimization [21], and joint mobility–manipulation modeling [6]. Xiaomi-Robotics-1 scales UMI pretraining beyond 100K hours before aligning the policy with robot embodiments and instructions [22]. LingBot-VLA 2.0 combines broader pretraining with whole-body action interfaces and predictive semantic and geometric supervision [23]. These systems broaden the range of tasks and embodiments a single policy can support, while leaving coordination between heterogeneous control channels an important architectural concern.

Beyond scale, architecture determines how perceptual features and action streams interact. VITRA concatenates an extracted VLM cognition feature with state and noisy action tokens in a DiT [24], while

Qwen-VLA jointly processes VLM hidden states and noisy actions through self-attention [25]. RLDX-1 extends this approach to modality-specific cognition, action, and optional physical-signal streams coupled through joint self-attention [26]. For coordination across body subsystems, AC-DiT conditions manipulation on a mobility prior [8], InCoM and GeoHAT introduce structured arm–base coordination [1, 2], and MoPA aligns perception separately for each subsystem [9]. DreamTrajectory guides whole-body actions with end-effector trajectories and refines candidates through a trajectory world model at test time [27]. For humanoids, ω -0 combines future observation embeddings with controller-compatible action latents for concurrent locomotion and manipulation [28]. MM-ABC applies joint attention [29] to specialized manipulation and mobility streams while keeping extracted VLM features as read-only context. The action streams retain separate transformations and asymmetric temporal visibility; we study how the action prediction target affects denoising and task allocation in this two-stream setting.

2.2 Representation Alignment and Predictive Supervision

Spatially grounded policies incorporate three-dimensional scene representations [30, 31], point-cloud conditioning [32, 33], and spatially informed VLA architectures [34, 35]. Further work models temporal structure [36, 37], learns dynamic-aware 3D representations for scalable robot learning [3], or integrates geometry with semantic features [4, 38–40]. The interface to the action expert also matters: policies use cross-attention [41, 42], final-layer features [43], or predictive embeddings [44]. DeepStack, Qwen3-VL, and DeepVision-VLA further motivate conditioning across network depth [45–47]. MM-ABC uses sparse intermediate VLM features as read-only perceptual context. This interface preserves multilevel visual information without requiring action tokens to modify the VLM token sequence.

Complementary approaches improve representations through teacher supervision, building on feature alignment and distillation [48, 49]. Spatial Forcing [50] and GLaD [51] supervise internal policy features with geometry; ROCKE extends alignment across layers [52], while VEGA targets the visual encoder [53]. Supervision ranges from object geometry [54, 55], depth [56], and attended-region reconstruction [57] to spatial and semantic knowledge [58–60], latent actions [61], and temporal geometry [62]. These methods differ both in the information supplied by the teacher and in the part of the policy receiving supervision, making the alignment pathway an important design choice.

Predictive supervision extends these ideas to future images [63], video embeddings [64], and latent world representations [5, 65, 66], including geometric evolution [67, 68]. PHR-VLA, WAM4D, and MECo-WAM use removable prediction branches to retain training benefits without deploying the auxiliary predictor [7, 69, 70]. MM-ABC similarly predicts future VGGT- Ω features during training [71]. Its future and action streams remain mutually masked, so geometric supervision shapes shared perceptual features without making future tokens policy inputs. The resulting design complements current-scene conditioning with a predictive learning signal, while removing the teacher and future branch at deployment.

2.3 Clean-Action Prediction and Cross-Stream Interaction

The prediction target is a longstanding design choice in diffusion and flow models [72–74], including Transformer-based generators [75]. In robotics, direct clean-action prediction already spans several settings. DP3 uses sample prediction for high-dimensional action generation [76], and RDT-1B trains a denoising network to regress clean action chunks for bimanual manipulation [77]. ManiCM adopts action-sample prediction within consistency distillation for one-step control [78]; FA-RDP likewise uses action-space reparameterization in distillation for reactive contact-rich manipulation [79]. More recently, ABot-M0 combines clean-action outputs with a velocity-space objective [43], while VGFM uses action-space predictions for intermediate value guidance in offline reinforcement learning [80].

Related image-generation studies provide complementary perspectives: JiT motivates clean-data prediction through manifold structure and finite capacity [81], Pixel MeanFlow separates clean outputs from velocity-based objectives [82], and MiniT2I explores a minimalist generation framework [83]. Across these settings,

network outputs, training objectives, and sampling procedures are separate design choices: clean-action outputs can be trained with velocity-space losses, and consistency distillation differs from ordinary denoising or flow training. Although endpoint and velocity predictions are algebraically convertible along an affine flow path, loss weighting and preconditioning can change their optimization behavior. Controlled robot-policy studies reinforce the importance of separating these effects from architecture and iterative prediction [84]. Building on established clean-action prediction, MM-ABC studies its role in coupled arm–base streams: how the parameterization affects the noise retained in endpoint estimates, denoising accuracy across noise levels, and the arm–body task allocation realized by few-step sampling. We evaluate these effects against an analytical Bayes-optimal denoiser and in full-policy ablations.

3 Understanding Clean-Action Prediction

Before introducing MM-ABC, we examine whether a two-stream action network should predict clean trajectories or flow velocities. Following JiT’s distinction between prediction and loss spaces [81], we compare both parameterizations under a common objective on a synthetic task whose action geometry and Bayes-optimal denoiser are known in closed form. The two variants share architecture, conditioning, data, noise, and loss, and differ only in what the network emits.

Parameterizations. For a clean chunk x and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, flow time $t \in [0, 1]$ defines

$$z_t = (1 - t)\epsilon + tx, \quad v = x - \epsilon. \quad (1)$$

An x -head estimates x directly, whereas a v -head estimates v . With shared conditioning c , the two networks produce the following endpoint estimates of the clean chunk:

$$\hat{x}^{(x)} = f_\theta(z_t, t, c), \quad \hat{x}^{(v)} = z_t + d(t)g_\theta(z_t, t, c), \quad (2)$$

where $d(t) = \max(1 - t, 0.05)$ stabilizes conversion near the clean endpoint. Both minimize masked endpoint MSE with inverse-square weights $d(t)^{-2}$, normalized to unit minibatch mean, which equals velocity regression for the v -head wherever $d(t) = 1 - t$. Both variants recover the sampling velocity as $(\hat{x} - z_t)/d(t)$.

Synthetic task. We generate paired manipulation and body chunks of shapes 16×58 and 16×22 online. A condition $c_0 \in \mathbb{R}^{12}$ maps to an eight-dimensional task vector τ , and an equiprobable mode $m \in \{0, 1\}$ sets the arm share $\alpha_m \in \{0.35, 0.65\}$:

$$\text{vec}(x_s) = a_{e,s}Q_{e,s}[\rho_s(m)\tau; b_s(2m - 1); u_s], \quad (3)$$

where $s \in \{M, B\}$, $\rho_M = \alpha_m$, $\rho_B = 1 - \alpha_m$, $b_M = 0.12$, $b_B = 1.60$, and $u_s \sim \mathcal{N}(0, I_4)$ is private variation. Each $Q_{e,s}$ has 13 orthonormal, temporally smooth columns on the valid channels of embodiment profile e , one of six profiles, two of which have no body supervision. Both networks receive $(z_{t,M}, z_{t,B}, t, c_0, e, m)$ with channel masks, so the arm–body allocation is given and the comparison isolates the prediction target. A four-block, width-512 transformer uses separate stream transformations, masked cross-stream attention, and no raw-input skip inside either decoder. Paired runs share initialization, batches, noise, and flow times, and train for 12,000 updates with batch size 256 and $t \sim \text{Beta}(1.5, 1)$; we report eight paired seeds, or sixteen seeds for the time-resolved metrics in panels (b) and (c).

(a) Target geometry. For one embodiment, we stack flattened 928-dimensional manipulation chunks into matrices for x , v , and ϵ , center them, and plot normalized singular values. The clean spectrum terminates at rank 13, as prescribed by the generator, while velocity and noise retain variation across the ambient space. A v -head must therefore reproduce noise directions that are absent from the clean-action subspace.

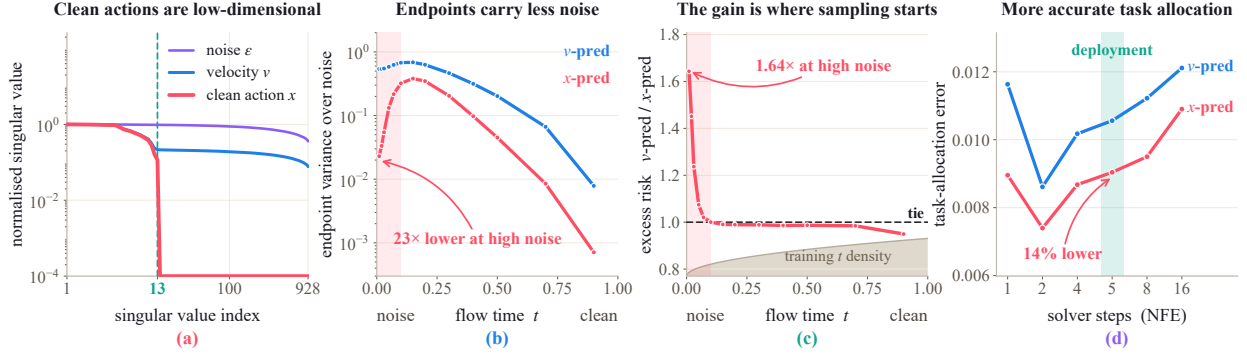


Figure 1. Clean-action versus velocity prediction under a common loss and a given arm-body allocation. (a) Clean chunks occupy a prescribed 13-dimensional subspace, whereas velocity and noise span the ambient space. (b) Variance of the endpoint estimate across noise draws with the clean chunk fixed. (c) Excess risk over the Bayes-optimal denoiser, as a v/x ratio. (d) Task-allocation error of sampled chunks against the number of Euler steps.

(b) Noise in the endpoint estimate. We compare both heads through their endpoint estimates $\hat{x}$, which live in the same space. Holding a clean chunk fixed, we draw 24 independent noises, compute $\hat{x}$ for each, and report its variance normalized by the clean-chunk variance. x -prediction yields lower variance at every tested flow time: 0.023 versus 0.54 at $t = 0.01$, a 23 $\times$ reduction, and 0.35 versus 0.63 at $t = 0.2$. Ridge probes on the final manipulation-token features show the same pattern internally: the injected noise is recoverable with held-out R^2 of 0.62 for x -prediction and 0.94 for v -prediction, against -0.09 for both with permuted rows. Because neither decoder has a raw-input skip, a v -head must carry this noise through its layers to form its endpoint estimate $z_t + d(t)g_\theta$ from the noisy input.

(c) Error beyond the Bayes-optimal denoiser. At fixed t , we compute endpoint MSE over valid entries and subtract the MSE of the Bayes-optimal conditional mean given the same inputs, available analytically in the known basis. This excess risk measures approximation error beyond the uncertainty inherent in denoising. The v/x ratio reaches 1.64 at $t = 0.01$, so x -prediction is markedly more accurate in the high-noise regime where every sampling trajectory begins.

(d) Sampled trajectories. We sample from Gaussian noise with uniform Euler steps and read each generated chunk out in the known latent coordinates. The task-allocation error is the distance between each stream’s realized share of τ and the nearest valid share, averaged over the two streams; it counts coordination errors even when the chunk stays inside the action subspace. x -prediction lowers this error at every tested budget from 1 to 16 steps, from 0.0106 to 0.0090 at five steps, the budget used by our policy. It also yields lower off-subspace mass at every tested budget (0.095 versus 0.104 at five steps). Its sliced Wasserstein distance to the true latent distribution, which also reflects the private variation u_s , is lower at 8 and 16 steps (0.075 versus 0.076 and 0.046 versus 0.050).

In this task, the allocation is supplied through m . In MM-ABC, it is inferred from observations, the instruction, and robot state, and the two action streams exchange it through masked joint attention (Section 4.3). The component ablation in Section 6.2 evaluates clean-action prediction in this learned setting, where it outperforms velocity prediction by 3.6 points on RoboCasa365 composite-seen tasks.

4 MM-ABC Architecture

4.1 Overview

At environment step n , MM-ABC receives current multi-view images I_n , a language instruction with embodiment and control metadata, and normalized robot state $s_n \in \mathbb{R}^{80}$ with validity mask $m_n^s \in \{0, 1\}^{80}$. It

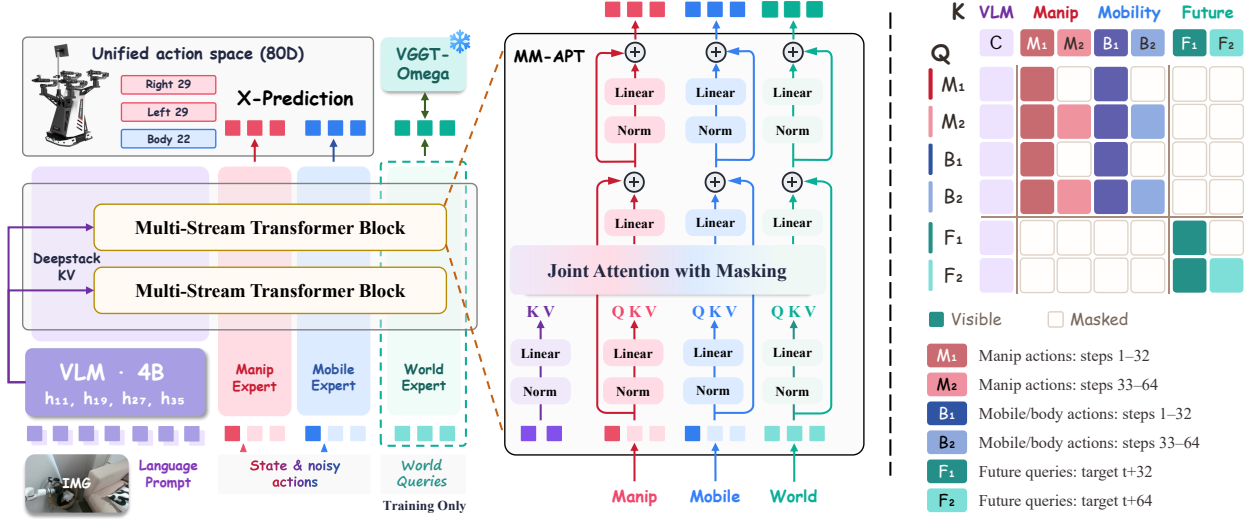


Figure 2. MM-ABC architecture. **Left:** sparse VLM features condition MM-APT; a training-only future branch receives geometric supervision from frozen VGGT-Omega features. **Middle:** joint transformer blocks (MM-JiT in the diagram) combine stream-specific transformations with masked attention and read-only perceptual context. **Right:** action streams communicate within each segment, and far tokens additionally read near tokens. Future queries follow the same temporal ordering but remain isolated from actions.

generates a 64-step normalized action chunk $X \in \mathbb{R}^{64 \times 80}$, split into manipulation $X_M \in \mathbb{R}^{64 \times 58}$ and body motion $X_B \in \mathbb{R}^{64 \times 22}$. Embodiment-specific masks identify supervised entries; native control mappings are described in Section 5. We distinguish environment step n from flow time $t \in [0, 1]$.

As shown in Figure 2, a Qwen3-VL-4B backbone encodes the current images and instruction. The **Mobile Manipulation Action Prediction Transformer (MM-APT)** reads this context and jointly denoises two action streams using the clean-endpoint interface of Section 3. A training-only future stream predicts later geometric features from the same context, while remaining isolated from action tokens.

4.2 Sparse Multilevel Perceptual Context

Mobile manipulation requires both recognizing the intended interaction and resolving local geometry as the viewpoint changes. Conditioning only on the final VLM layer places both demands on a single representation optimized for the backbone’s output tasks. Features at intermediate depths offer additional access to visual information that may be less explicit in the final representation, motivating conditioning across depth [45, 47]. At the other extreme, separately attending to every VLM layer would introduce many feature interfaces for the action expert to reconcile during training. We instead retain the final representation as a common semantic context and add a small number of intermediate features through gated residual updates.

We extract hidden states $H^l \in \mathbb{R}^{S \times 2560}$ from zero-indexed VLM layers $l \in \{11, 19, 27, 35\}$, where S is the image–text sequence length. Starting from $C_{-1} = H^{35}$, we inject intermediate features into the first three of the 16 expert blocks as gated residual updates:

$$C_i = C_{i-1} + g_i \odot P_i(\text{LN}(H^{l_i})), \quad i = 0, 1, 2. \quad (4)$$

Here $(l_0, l_1, l_2) = (11, 19, 27)$, LN denotes layer normalization, P_i is a tokenwise linear map preserving width 2560, and $g_i \in \mathbb{R}^{2560}$ is a zero-initialized gate; $\odot$ denotes elementwise multiplication broadcast over tokens. Later blocks retain $C_i = C_{i-1}$. Each block projects its context into width-1024 keys and values, which action tokens read without updating the context sequence. Gradients still reach the VLM, and context projections are reused across noise draws and sampling steps. The zero-initialized gates make the initial interface identical to final-layer conditioning, allowing intermediate features to enter gradually as their gates are learned. Sparse, cumulative injection limits changes in the context source across expert depth, and the

read-only design holds perceptual evidence fixed throughout each denoising trajectory. These choices provide a controlled route to multilevel information without requiring a separate connection to every backbone layer.

4.3 Structured Two-Stream Action Generation

Manipulation and body motion contribute differently to the same task: the body changes reachability and viewpoint, while the arm controls local interaction. Separate transformations accommodate these different control semantics, but independent predictors would have to infer their partner’s motion indirectly. Joint attention lets each stream adjust its prediction using the other’s evolving action representation, while both remain grounded in the same observation and full robot state.

Token construction. MM-APT uses 16 transformer blocks of width 1024, with 16 attention heads and feed-forward width 4096. Separate stream encoders project the noisy actions $Z_{t,s}$, concatenate a broadcast sinusoidal time embedding, and fuse them through a multilayer perceptron, for $s \in \{M, B\}$. Each stream prepends a state token encoded from the full $[s_n; m_n^s] \in \mathbb{R}^{160}$, yielding 65 tokens. Learned position and segment embeddings distinguish action steps and divide the chunk into near (state token and first 32 actions) and far (remaining 32 actions) segments.

Structured communication. Streams retain separate normalization, attention projections, and SwiGLU feed-forward layers, but their tokens participate in joint masked attention with read-only perceptual context. Manipulation and body tokens communicate bidirectionally within each segment. Far queries additionally read near keys, whereas near queries cannot read far keys, allowing the tail to build on the immediate plan without influencing it. This asymmetry reflects the unequal roles of the two horizons: the prefix must support the next physical interaction, while the tail anticipates states that will be observed again before execution. For example, an immediate reach should be grounded in the current object and base configuration, while a later repositioning can be refined after new observations arrive. The distinction between immediate execution and longer-horizon organization also appears in accounts of hierarchical motor control [85]. Both horizons supervise the shared parameters under this near-to-far attention pattern.

Future queries obey the same temporal ordering but cannot exchange information with either action stream. All streams read valid context tokens; padding and inactive body streams are masked. After the final block, separate normalized multilayer decoders discard the state tokens and predict clean chunks $\hat{X}_M$ and $\hat{X}_B$, without a direct noisy-input skip.

4.4 Future Geometry as Auxiliary Supervision

Current-scene understanding alone does not specify which spatial relations matter for the next interaction. A robot approaching a handle, for example, needs features informative about how the handle and end effector will come into alignment, as well as the handle’s present appearance. This motivates training the VLM representation to support anticipation of task-relevant geometry. Predicting future geometric features provides spatially distributed supervision beyond the action vector: successful prediction requires preserving information about the scene that helps explain its subsequent configuration. Using features from a geometry-oriented teacher [71] directs this objective toward spatial structure, while future rather than current targets encourage the representation to capture how that structure evolves during the demonstrated task.

The training-only World Expert predicts geometric features at environment steps $n + 32$ and $n + 64$ from current perceptual context. Each horizon has 192 learned width-1024 queries, corresponding to an 8×8 grid for each of up to three camera views. These queries form a separate stream in the joint transformer and receive neither action tokens nor future images.

A frozen VGGT-Omega teacher processes recorded future images separately at each horizon and pools its features to the same spatial grids. A layer-normalized linear decoder maps future-query outputs to

2048-dimensional predictions $\hat{F}_{brj}$ matching teacher targets F_{brj}^* , where b, r , and j index examples, horizons, and view-spatial tokens. The alignment objective is

$$\mathcal{L}_{\text{future}} = \frac{\sum_{b,r,j} q_{brj} \left[1 - \cos_{\delta}(\hat{F}_{brj}, F_{brj}^*) \right]}{\max(1, \sum_{b,r,j} q_{brj})}. \quad (5)$$

Here $q_{brj} \in \{0, 1\}$ masks unavailable future frames, views, and invalid teacher vectors; teacher targets receive no gradients. The stabilized cosine is $\cos_{\delta}(u, v) = u^T v / [\sqrt{\|u\|_2^2 + \delta} \sqrt{\|v\|_2^2 + \delta}]$ for a small $\delta > 0$. This auxiliary predictor supervises the shared perceptual representation to anticipate demonstrated geometry. Because the future stream reads the same VLM features as the action streams, its loss trains the backbone to make this predictive geometric information available to the policy. Future observations define training targets, while action generation reads the current perceptual context under the mutual attention mask. Both the future stream and teacher are removed at inference.

4.5 Training Objective

We construct $Z_t = (1 - t)E + tX$ using independent Gaussian noise E and $t \sim \text{Beta}(1.5, 1)$ clipped at 0.999, and predict $\hat{X}$ from current context and state. For a supervised minibatch of size B , let $M_{b,s}$ mask valid action entries and $\mathcal{S}_b$ contain the active streams of example b . Using $d(t)$ from Section 3, we define the normalized weight $\bar{w}_b = d(t_b)^{-2} / [B^{-1} \sum_{b'=1}^B d(t_{b'})^{-2}]$. The masked action loss is

$$\mathcal{L}_{\text{action}} = \frac{1}{B} \sum_{b=1}^B \frac{\bar{w}_b}{|\mathcal{S}_b|} \sum_{s \in \mathcal{S}_b} \frac{\|M_{b,s} \odot (\hat{X}_{b,s} - X_{b,s})\|_F^2}{\|M_{b,s}\|_1}. \quad (6)$$

The squared Frobenius norm $\|\cdot\|_F^2$ sums squared errors, and $\|M_{b,s}\|_1$ counts valid entries, giving active streams equal weight regardless of their supervised dimensions. We average this loss over four independent noise/time draws sharing one context computation and evaluate future supervision once:

$$\mathcal{L} = 1.0 \mathcal{L}_{\text{action}} + 0.05 \mathcal{L}_{\text{future}}. \quad (7)$$

At inference, the action streams generate chunks from Gaussian noise with five Euler steps, using the endpoint-to-velocity conversion in Section 3; the robot executes a prefix before replanning.

5 Multi-Embodiment Pretraining Data Engine

Robot datasets differ in control semantics, coordinate conventions, embodiment, and sampling frequency. We construct MM-ABC’s pretraining corpus by auditing each source, standardizing its state and action representations, and retaining temporally contiguous valid trajectories in a shared masked interface. A balanced sampling scheme controls the contribution of each source to training.

After processing, our corpus contains **5,166.2 hours** and **400K+ episodes** of demonstrations from **12 datasets**, **51 subsets**, and **17 embodiments**, together with **65K+ unique natural-language instructions**. Real-world and simulated demonstrations account for 80% and 20% of the total duration, respectively. Platforms capable of mobile manipulation contribute approximately 61% of the cleaned hours. Within this group, 1,371.7 hours (26.6% of the full corpus) both retain base-control channels and exhibit nontrivial base motion. Our self-collected mobile manipulation set contains **30+ hours** of demonstrations across **40+ tasks**, with all cleaned demonstrations included in pretraining (Section 5.2).

5.1 Corpus Composition and Training Mixture

Figure 3 summarizes the corpus and its *training sampling probabilities*; Figure 4(a) reports the duration contributed by each dataset. These quantities differ because training uses a rebalanced mixture. Some sources contribute 700–900 cleaned hours or more, while smaller sources add embodiments and control configurations with limited representation in the corpus.

Data sources. The corpus combines eleven public datasets with our MM-30 collection. BEHAVIOR-1K and InternData-A1 are simulated; the other sources are recorded on physical robots or, for Hy-Embodied 0.5, with a handheld UMI device.

- **ABC-130k** [86]: a bimanual teleoperation dataset of 134,806 episodes and 3,553 hours, collected on low-cost stations with two 6-DoF YAM arms. Its 195 tasks cover pick-and-place, folding, handover, insertion, tool use, and assembly.
- **RoboCOIN** [87]: over 180K teleoperated bimanual demonstrations from 15 robotic platforms, spanning 421 tasks in 16 residential, commercial, and working scenarios, with hierarchical annotations ranging from trajectory-level concepts to frame-level kinematics.
- **BEHAVIOR-1K** [88]: a simulation benchmark of 1,000 everyday household activities built on OmniGibson. Its 2026 challenge release provides 20,000 teleoperated demonstrations of 100 long-horizon tasks on the wheeled bimanual Galaxea R1-Pro.
- **Hy-Embodied 0.5** [89]: bimanual demonstrations collected with a fingertip UMI device tracked by optical motion capture and recorded from head and wrist views. The full corpus exceeds 10,000 hours, and the public release provides 2,163 hours over 70+ tasks.
- **AgiBotWorld EE** [90]: AgiBot World contains 1,001,552 trajectories and 2,976.4 hours over 217 tasks, 87 skills, and 106 scenes, collected by more than 100 AgiBot robots in five deployment domains.
- **InternData-A1** [91]: synthetic data from a compositional simulation pipeline, with over 630K trajectories and 7,433 hours across 70 tasks, 18 skills, and 227 scenes, covering rigid, articulated, deformable, and fluid objects on Franka Panda, AgileX Split Aloha, ARX Lift-2, and AgiBot Genie-1.
- **RealSource World** [92]: 11,428 episodes of long-horizon manipulation over 35 tasks on the RS-02 dual-arm humanoid, recorded in kitchens, conference rooms, convenience stores, homes, and industrial settings with atomic-skill segments and per-episode quality assessments.
- **Galaxea Open-World** [93]: 500+ hours of real-world mobile manipulation over 150+ tasks in 50 scenes, collected on a single embodiment, the Galaxea R1-Lite with two 6-DoF arms, a 3-DoF torso, and an omnidirectional base, and annotated with bilingual subtask labels.
- **DROID** [94]: in-the-wild single-arm Franka Panda demonstrations comprising 76K trajectories and 350 hours across 564 scenes and 86 tasks.
- **AgiBotWorld 2026** [95]: real-world data collected on the AgiBot G2 platform in a free-form mode, covering commercial spaces such as retail stores as well as home scenarios.
- **HIW-500** [96]: 500+ hours and 23K+ episodes of whole-body teleoperation of Unitree G1 humanoids in 12 real homes in Southeast Asia, covering 10+ household tasks with 161 subtask labels.
- **MM-30 (ours)**: 30+ hours of coordinated base and dual-arm demonstrations over 40+ tasks on our mobile manipulation platform (Section 5.2).

Balanced sampling. Sampling weights are assigned to source-specific training profiles according to their cleaned duration. For a profile i containing h_i hours, we apply a sublinear duration weighting, followed by a cap on each profile’s normalized sampling probability:

$$\tilde{p}_i = h_i^{0.4}, \quad p_i = \text{CapNorm}(\tilde{p}_i; 0.2). \quad (8)$$

CapNorm normalizes the weights, limits each profile’s probability to 0.2, and redistributes excess probability mass among uncapped profiles. This reduces the concentration of sampling probability in the largest profiles. Figure 3 aggregates these probabilities across profiles belonging to the same dataset.

5.2 Self-Collected Mobile Manipulation Dataset

To complement public datasets, we collect **MM-30**, a real-world mobile manipulation dataset with coordinated base and dual-arm motion. Figure 5 summarizes its tasks, objects, and scenes.

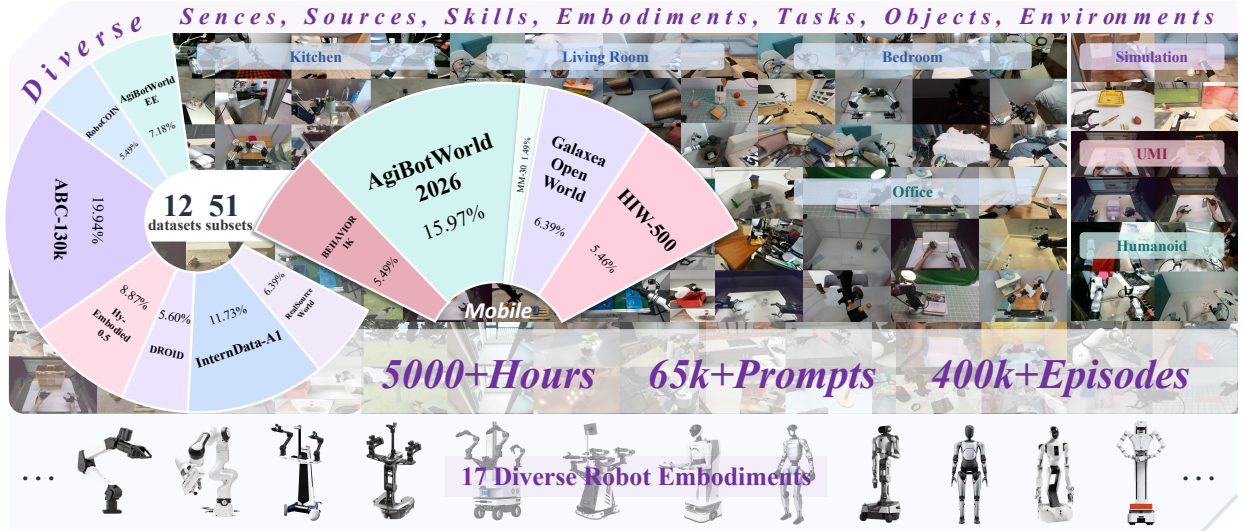


Figure 3. Overview of the MM-ABC multi-embodiment pretraining corpus. The corpus contains 5,166.2 cleaned hours from 12 datasets and 51 subsets, spanning 17 embodiments. Sector areas show training sampling probabilities aggregated by dataset; the surrounding panels illustrate environments and embodiments.

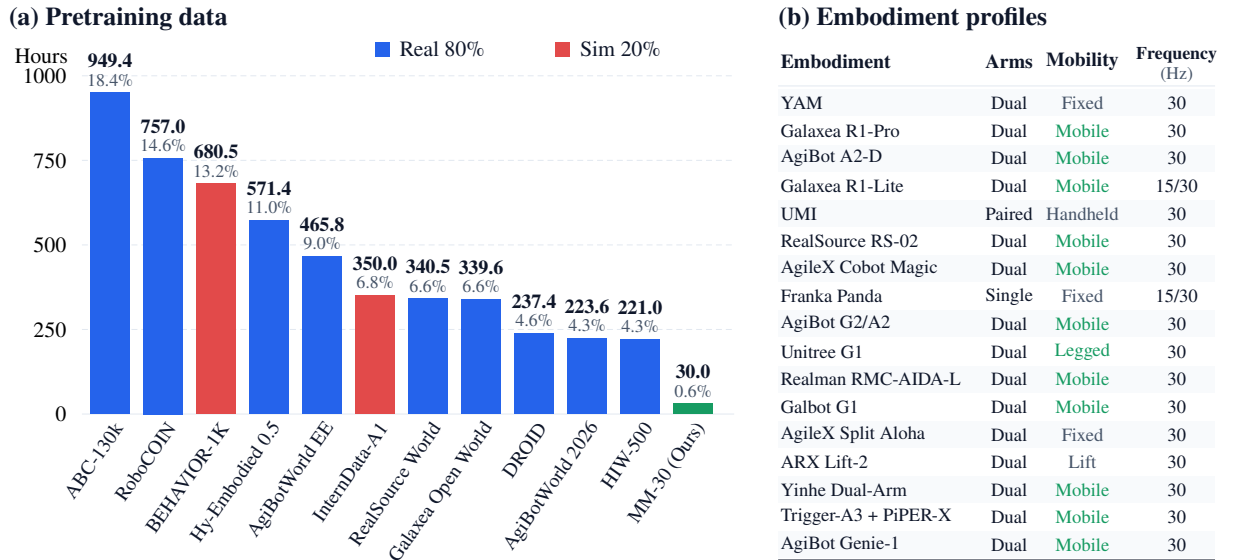


Figure 4. Pretraining data and embodiment composition. (a) Dataset hours and duration shares. (b) Arm configuration, mobility, and control frequency.

Hardware platform. The platform comprises a **HexFellow Trigger-A3** omnidirectional mobile base and two **AgileX PiPER-X** 6-DoF manipulators. The base repositions the robot in the horizontal plane, and the two arms perform object interactions, including bimanual handovers. Their active control channels map to the base and left- and right-arm slots of the 80D interface (Figure 4(b)).

Task coverage and diversity. We collect **30+ hours** of teleoperated demonstrations spanning **40+** mobile manipulation tasks. Multi-view demonstrations benefit manipulation learning beyond viewpoint generalization [97]; we therefore record each task from multiple viewpoints and vary initial configurations, object layouts, distractors, and scene appearance to broaden visual and spatial coverage. The demonstrations include picking, placing, opening, closing, pouring, wiping, fetching, and short-horizon rearrangement, with base motion adjusting the reachable workspace during manipulation.

MM-30 A real-world mobile manipulation dataset

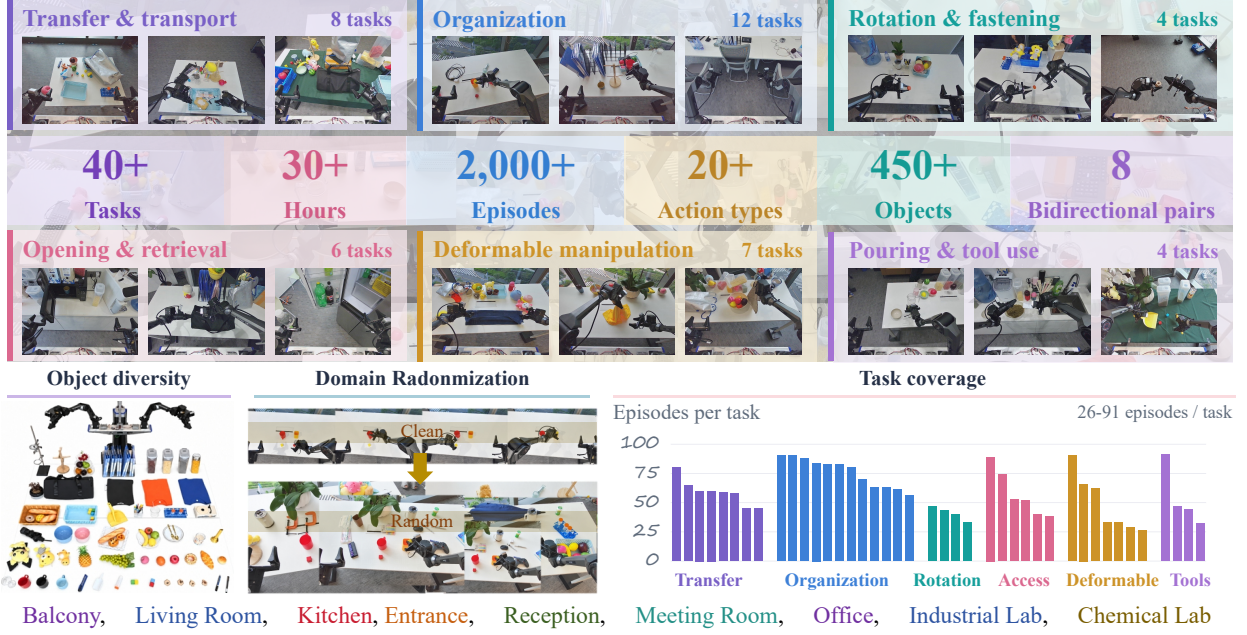


Figure 5. MM-30: self-collected mobile manipulation dataset. Representative tasks, object and scene diversity, and episode counts per task.

Role in pretraining. After the semantic audit, coordinate alignment, and quality filtering described in Section 5.4, all cleaned demonstrations from MM-30 are included in MM-ABC pretraining (Figure 4(a)).

5.3 Embodiment Diversity and Unified Control Interface

The corpus spans 17 embodiments, including single- and dual-arm systems, fixed-base manipulators, wheeled mobile manipulators, a full-body humanoid, lift-equipped platforms, and handheld UMI demonstrations. The number of valid stored dimensions ranges from 10 to 43, and control frequencies range from 15 to 30 Hz. Figure 4(b) summarizes their arm configurations, mobility, and control frequencies.

For joint training across embodiments with different sensors and control channels, MM-ABC maps their state and action fields into a fixed **80D canonical state/action interface**:

$$\mathbf{a}_t = \mathbf{a}_t^L \oplus \mathbf{a}_t^R \oplus \mathbf{a}_t^B \in \mathbb{R}^{29+29+22}, \quad \mathbf{m}_t \in \{0, 1\}^{80}, \quad (9)$$

Here $\oplus$ denotes concatenation and $\mathbf{m}_t$ is an embodiment-specific validity mask. The left and right 29D manipulation blocks reserve slots for arm joints, 3D end-effector (EEF) position, 6D rotation representations, gripper state, and optional hand joints. The 22D body block reserves slots for mobile-base motion, torso, lift, head, and auxiliary controls. Only channels supported by the verified source schema are populated. Unavailable channels are set to zero and masked out of the action loss.

State and action use the same canonical field layout in the processed corpus. When a source stores both joint and EEF representations, both can occupy the 80D vector; the source’s control mode determines which representation provides action supervision. They are not treated as simultaneous, independent action targets.

5.4 Data Engine: Semantic Audit, Cleaning, and Canonicalization

The pipeline in Figure 6 standardizes field semantics before applying common quality checks. State and action tensors can encode joint targets, Cartesian poses, deltas, velocities, or delayed controller commands. Sources also differ in parent coordinate frames, tool center points (TCPs), rotation conventions, and gripper conventions, requiring explicit interpretation before conversion.

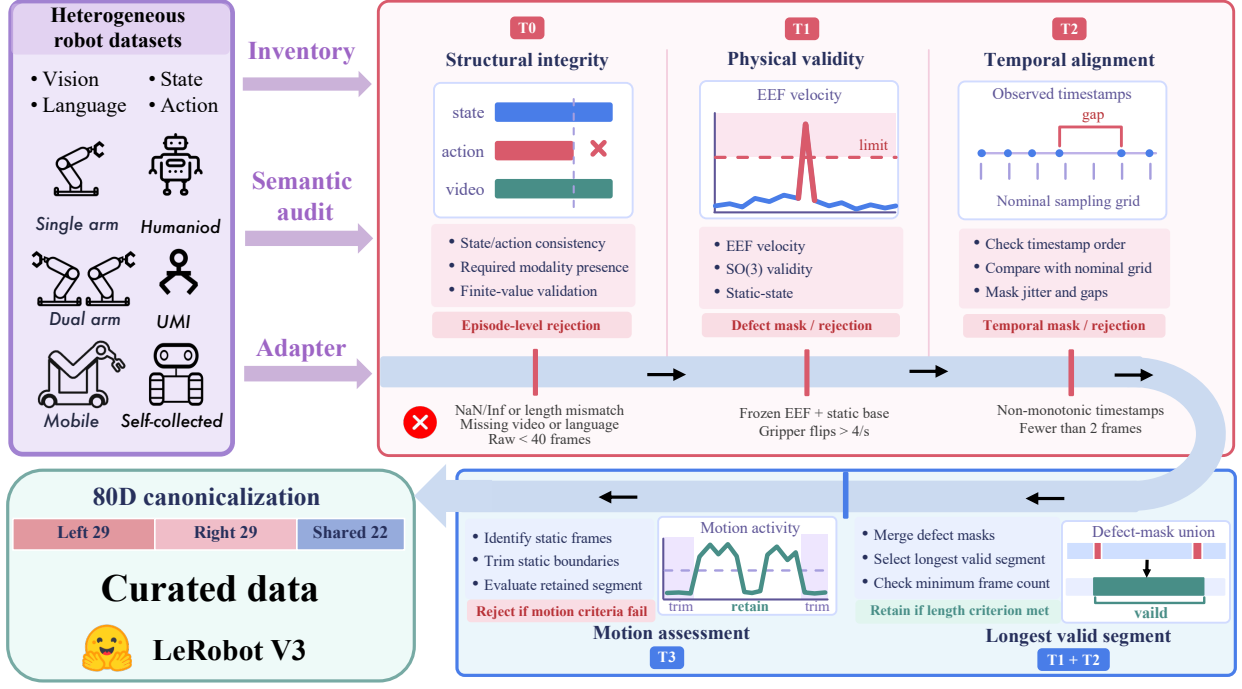


Figure 6. MM-ABC data engine. Sources undergo semantic auditing, conversion with dataset-specific adapters, and four stages of quality filtering. Retained contiguous segments are encoded in the masked 80D interface, independently validated, and incorporated into the training mixture.

Source audit and field semantics. For each source, we fix its revision and record its schema, camera streams, control frequency, and robot metadata. A dataset-specific *frame contract* specifies the physical meaning of each state/action field: units, action type, parent frame, TCP, rotation convention, gripper direction, and temporal semantics. Dataset-specific adapters then convert units, unwrap Euler angles where needed, reconstruct rotations in SO(3), and align coordinate frames and TCPs. Explicit field semantics distinguish velocity commands from pose targets and identify actions expressed relative to different tool centers, parent frames, or rotation conventions.

Staged quality filtering. Filtering proceeds from inexpensive structural checks to more expensive motion analysis. **Tier 0** checks structural integrity, rejecting trajectories with insufficient length, state/action length mismatches, non-finite values, missing required videos, or missing language annotations. **Tier 1** checks physical validity, including implausible EEf speed, workspace violations, invalid rotations, frozen signals, and excessive gripper switching. **Tier 2** checks timestamp monotonicity and flags temporal jitter and large gaps. We merge the Tier 1–2 defect masks and retain the *longest contiguous valid segment*. Finally, **Tier 3** trims stationary prefixes and suffixes and rejects trajectories dominated by stationary behavior or negligible displacement of the end effector and base.

Filtering preserves temporal continuity: invalid transitions delimit candidate segments, and *interior frames are never removed from a retained segment*. This avoids introducing unrecorded time jumps into action chunks. A two-pass implementation first records all retention, rejection, and trimming decisions in a manifest, then writes the retained segments in the canonical format.

Independent validation. A separate validator checks the serialized data for schema consistency, finite values, timestamp continuity, video seeking and decoding, rotation orthogonality and round-trip consistency, compliance with the state/action frame contracts, and correct validity masks and normalization statistics. Corpus statistics and normalization parameters are recomputed from the cleaned data after serialization.



Figure 7. Simulation benchmarks. Representative task scenes from LIBERO/LIBERO-Plus, RoboCasa365, EBench, and ManiSkill-HAB, covering fixed-base and mobile manipulation.

6 Experiments

We organize our evaluation around three questions. First, how does MM-ABC compare with strong VLA and WAM baselines on mobile and fixed-base manipulation, including under distribution shifts? Second, how do sparse multilevel perceptual conditioning, clean-action prediction, and future geometric supervision contribute to performance? Third, how effectively does MM-ABC coordinate base motion and object interaction on a physical robot? To address these questions, we evaluate MM-ABC on EBench, RoboCasa365, ManiSkill-HAB, LIBERO, and LIBERO-Plus, conduct controlled component ablations, and compare policies on five real-world mobile manipulation tasks. We also integrated our implementation into XPolicyLab [98].

6.1 Simulation Experiments

Benchmarks and protocol. Figure 7 illustrates representative tasks from the simulation benchmarks. We follow the official task splits and success criteria of each benchmark. Starting from the pretrained MM-ABC weights, we adapt separate policies using the demonstrations available for each benchmark or task suite. The comparisons include visuomotor policies, vision–language–action (VLA) models, and world-action models (WAMs). Observation modalities and training budgets are given with each benchmark. The benchmark comparisons assess the full policy, while Section 6.2 examines individual components under a common from-scratch training protocol on the RoboCasa365 composite-seen tasks.

EBench. EBench comprises 26 indoor tasks spanning mobile pick-and-place, long-horizon mobile manipulation, and dexterous tabletop manipulation [99]. Its tasks vary in scene, skill, horizon, precision, and operating mode, assessing both workspace repositioning and precise object interaction. A single checkpoint is evaluated on the held-out test split following the official evaluation protocol, using task success and a stage-wise progress score. We post-train MM-ABC for 100k steps with a batch size of 512.

As shown in Figure 8, MM-ABC achieves a success rate of **44.71%** and a progress score of **59**. The strongest baseline, $\pi_{0.5}$, achieves 41.41% and 54, respectively. The corresponding improvements are 3.30 percentage points in success rate and 5 points in progress score, indicating gains in both task completion and intermediate progress. This joint improvement is consistent with MM-APT’s design, which conditions manipulation and body actions on shared multilevel context while accommodating both mobile and fixed-base control.

RoboCasa365. The RoboCasa365 target evaluation comprises 50 household tasks in held-out kitchens: 18 atomic-seen, 16 composite-seen, and 16 composite-unseen tasks [105]. Atomic tasks emphasize individual interactions with objects and articulated fixtures, whereas composite tasks combine successive interactions with navigation. The seen/unseen labels refer to task inclusion in RoboCasa365’s predefined pretraining set. We use the full set of target demonstrations for post-training, with 120k steps and a batch size of 512.

Table 1 reports success rates for the three splits and their task-weighted average. MM-ABC achieves 78.4%, 52.7%, and 50.3%, respectively, with an overall average of 61.2%. These results exceed ABot-M0.5 by 7.8, 8.4, and 4.7 percentage points on the individual splits and by 7.0 points overall. The improvements extend

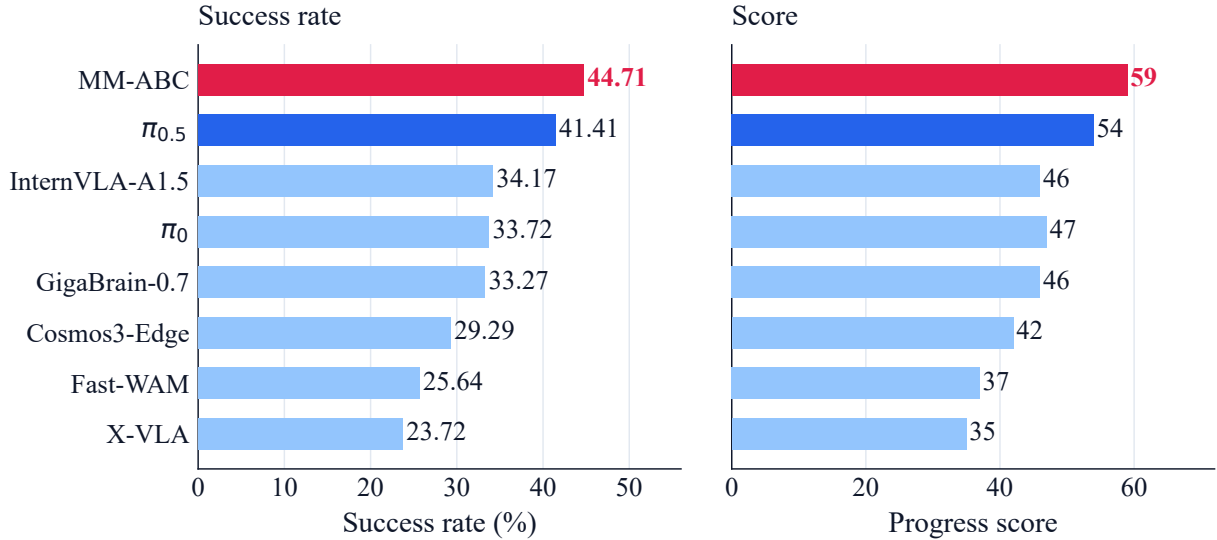


Figure 8. EBench evaluation. Success rate (left) and stage-wise progress score (right) for MM-ABC and selected baselines [19, 20, 100–104]. Higher is better.

Table 1. RoboCasa365 target evaluation with full demonstrations. Success rate (%). S/U denote seen/unseen tasks; the average is weighted by the split sizes (18/16/16). Best: shaded; second best: bold.

Method	Atomic-S	Composite-S	Composite-U	Average
GR00T-N1.5 [41]	60.6	35.0	33.3	43.7
Fast-WAM [103]	59.1	36.4	33.2	43.5
LingBot-VA [106]	63.5	37.3	32.1	45.1
ABot-M0.5 [6]	70.6	44.3	45.6	54.2
MM-ABC (Ours)	78.4	52.7	50.3	61.2

across atomic and composite tasks, with the largest margin on the composite-seen split. Composite tasks chain several interactions with navigation and require the base and arms to act in concert as reachability and viewpoint change. Across all compared methods, success rates are lower on composite tasks than on atomic tasks, which confirms composite tasks as the harder regime.

ManiSkill-HAB. ManiSkill-HAB evaluates low-level mobile manipulation with a Fetch robot, including physically simulated grasping and interaction with articulated objects [107]. SetTable requires retrieving a bowl from a drawer and an apple from a refrigerator and placing them on a table; we evaluate seven skills covering picking, placement, refrigerator opening, and drawer opening and closing. TidyHouse rearranges objects among open receptacles, whereas PrepareGroceries transfers objects between a refrigerator and a counter; both cover picking and placement across nine object categories. One policy is trained per suite for 100k steps with a batch size of 64, using head- and wrist-camera RGB images, proprioception, and language instructions as observations for every control step.

Tables 2 and 3 show that MM-ABC achieves the highest mean success rate on SetTable (86.4%), TidyHouse (70.2%), and PrepareGroceries (65.6%). Improvements on TidyHouse and PrepareGroceries are concentrated in picking. This pattern is consistent with MM-APT’s multilevel perceptual conditioning and coupled manipulation–body prediction, which target object acquisition across varied geometries.

LIBERO and LIBERO-Plus. We evaluate LIBERO on four fixed-base manipulation suites—Spatial, Object, Goal, and Long—with ten tasks per suite [111]. Spatial, Object, and Goal vary object arrangements, object identities, and task objectives, respectively, while Long emphasizes extended sequences of interactions. We use the standard LIBERO demonstrations for post-training and an evaluation budget of 50 trials per task. LIBERO-Plus introduces perturbations to camera configuration, robot initial state, language, lighting,

Table 2. ManiSkill-HAB SetTable [9]. Skill and mean success rates (%). * denotes depth input; – indicates an unavailable result. AnchorVLA’s mean covers six available skills. Best: shaded; second best: bold.

Method	Pick Apple	Pick Bowl	Place Apple	Place Bowl	Open Fridge	Open Drawer	Close Drawer	Mean
DP3* [76]	0.0	20.0	31.0	32.0	0.0	0.0	68.0	21.6
ACT [108]	28.0	28.0	8.7	13.0	2.0	0.0	85.7	23.6
DP [18]	21.3	20.7	28.0	69.3	7.3	0.0	55.0	28.8
RDT-1B [77]	12.0	10.7	32.0	18.7	82.7	44.0	100.0	42.9
AC-DiT* [8]	33.3	36.0	33.3	17.3	90.7	81.3	97.3	55.6
π_0 [19]	26.6	26.6	48.9	56.8	90.5	75.5	88.7	59.1
AnchorVLA [109]	22.7	44.5	64.3	63.8	88.9	–	100.0	64.0
MobileWAM [110]	46.0	46.0	63.7	64.7	99.3	91.0	100.0	73.0
GeoHAT* [2]	82.3	69.3	60.0	78.0	83.7	86.0	95.3	79.2
InCoM* [1]	59.4	84.1	84.1	82.5	87.3	88.9	100.0	83.8
MM-ABC (Ours)	81.3	86.7	71.0	81.3	96.7	92.0	95.7	86.4

Table 3. ManiSkill-HAB TidyHouse and PrepareGroceries [9]. Pick and Place average success rates (%) over nine object categories; Mean averages both groups. Best: shaded; second best: bold.

Method	TidyHouse			PrepareGroceries		
	Pick	Place	Mean	Pick	Place	Mean
ACT [108]	2.2	31.6	16.9	2.0	27.5	14.8
DP [18]	0.0	30.3	15.2	0.4	17.7	9.1
DP3 [76]	0.0	61.0	30.5	0.0	31.3	15.7
InCoM [1]	16.7	78.9	47.8	15.0	65.9	40.5
GeoHAT [2]	30.3	73.3	51.8	19.7	60.7	40.2
MM-ABC (Ours)	71.5	68.8	70.2	64.4	66.8	65.6

background, sensor noise, and scene layout, with 10,030 evaluation episodes in total [112]. For post-training, MM-ABC uses only the original LIBERO demonstrations, without additional training on perturbed demonstrations or environments.

As shown in Table 4, MM-ABC achieves 99.1% mean success on LIBERO, 0.5 percentage points above the strongest baseline, ABot-M0. It achieves the highest success rate on Spatial, ties for the highest on Object, and is within 0.4 and 0.1 points of the best results on Goal and Long, respectively. With the inactive body stream masked (Section 4.3), these results show that the shared perceptual representation and clean-action decoder also support fine-grained fixed-base manipulation.

Table 5 evaluates the LIBERO-trained MM-ABC policy on LIBERO-Plus without further training. MM-ABC achieves a total success rate of 82.8%, the highest among the compared methods. It exceeds Cosmos-Policy (82.2%) and ABot-M0 (80.5%), which are also trained only on the original demonstrations, as well as OpenVLA-OFT+ (79.6%) and GR00T-N1.6+ (79.4%), which are additionally trained on perturbed demonstrations. MM-ABC also achieves the highest language (88.9%) and background (96.1%) scores, consistent with conditioning actions on shared multilevel visual–language features. Its lowest category scores occur under camera (73.6%) and robot initial-state (66.5%) perturbations.

6.2 Ablation Studies

Experimental setup. We examine the model components on the 16 RoboCasa365 composite-seen tasks. All variants omit pretraining on our robot-data mixture and use a common training budget of 120k steps with a batch size of 64. This separates component comparisons from the pretraining in Table 1.

Table 4. LIBERO. Success rate (%) with 50 trials per task. Baselines include suite-specific and shared-policy configurations. Best: shaded; second best: bold.

Method	Spatial	Object	Goal	Long	Average
OpenVLA [17]	84.7	88.4	79.2	53.7	76.5
WorldVLA [113]	87.6	96.2	83.4	60.0	81.8
π_0 -FAST [114]	96.4	96.8	88.6	60.2	85.5
NORA [115]	92.2	95.4	89.4	74.6	87.9
π_0 [19]	96.8	98.8	95.8	85.2	94.2
UniVLA [116]	96.5	96.8	95.6	92.0	95.2
$\pi_{0.5}$ [20]	98.8	98.2	98.0	92.4	96.9
OpenVLA-OFT [117]	97.6	98.4	97.9	94.5	97.1
GR00T-N1.6 [41]	99.3	99.2	98.4	92.9	97.5
Fast-WAM [103]	98.2	100.0	97.0	95.2	97.6
StarVLA [118]	98.7	99.7	98.6	94.2	97.8
X-VLA [104]	98.2	98.6	97.8	97.6	98.1
Cosmos-Policy [119]	98.1	100.0	98.2	97.6	98.5
LingBot-VA [106]	98.5	99.6	97.2	98.5	98.5
ABot-M0 [43]	98.8	99.8	99.0	96.6	98.6
MM-ABC (Ours)	99.4	100.0	98.6	98.4	99.1

Table 5. LIBERO-Plus. Success rate (%) across seven perturbation categories [112]. Total is the success rate over all 10,030 evaluation episodes, following the official protocol. + denotes additional training on LIBERO-Plus perturbed demonstrations. Rankings span both groups. Best: shaded; second best: bold.

Method	Camera	Robot	Language	Light	Background	Noise	Layout	Total
<i>Additional training on perturbed demonstrations</i>								
π_0 + [19]	79.6	21.1	72.5	84.7	86.2	68.3	69.4	67.4
GR00T-N1.6+ [41]	92.6	33.5	80.1	93.6	95.4	93.6	75.0	79.4
OpenVLA-OFT+ [112]	92.8	30.3	85.8	94.9	93.9	89.3	77.6	79.6
<i>Trained on original LIBERO only</i>								
OpenVLA [17]	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
GR00T-N1.6 [41]	20.9	40.2	35.0	65.4	76.3	27.8	51.3	42.8
UniVLA [116]	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
Fast-WAM [103]	16.4	44.5	68.9	78.2	53.7	37.7	60.7	50.0
π_0 [19]	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
π_0 -FAST [114]	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
LingBot-VA [106]	40.9	83.0	86.4	82.3	53.1	64.4	76.2	69.5
OpenVLA-OFT [117]	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
X-VLA [104]	23.4	89.7	75.7	88.2	96.0	62.7	71.8	70.5
StarVLA [118]	52.5	49.8	88.5	95.7	95.7	73.0	76.9	74.1
ABot-M0 [43]	60.4	67.9	86.4	96.2	91.6	86.4	82.6	80.5
Cosmos-Policy [119]	75.8	63.3	81.7	96.5	88.9	92.7	82.2	82.2
MM-ABC (Ours)	73.6	66.5	88.9	94.9	96.1	86.4	80.5	82.8

Model variants. We compare five configurations. **Interleaved** conditions alternating action layers on VLM features, replacing the sparse multilevel conditioning used by the full model. **Last layer** removes DeepStack and conditions only on the final VLM representation. **x-pred** retains clean-action prediction but removes the future supervision loss. **v-pred** retains the future branch but replaces clean-action prediction with velocity prediction. The full **MM-ABC** configuration combines clean-action prediction, future geometric supervision, and DeepStack. Table 6 reports per-task success rates and their unweighted mean.

Action prediction and future supervision. The full model achieves the highest average success rate, 32.8%, and the highest success on 9 of the 16 tasks. Replacing x-prediction with v-prediction while retaining

Table 6. Component ablations on RoboCasa365 composite-seen tasks. Success rate (%) without robot-data pretraining. Average is the unweighted mean across 16 tasks. Rankings are computed within each row, across the five variants. Best: shaded; second best: bold.

Task	Interleaved	Last layer	x -pred	v -pred	MM-ABC
DeliverStraw	0.0	0.5	0.5	0.5	0.0
GetToastedBread	0.0	1.0	0.0	1.0	0.5
KettleBoiling	33.0	29.5	35.5	41.5	49.0
LoadDishwasher	18.5	28.0	22.0	27.0	33.0
PackIdentical	0.5	1.5	1.5	7.0	6.0
PreSoakPan	29.5	50.5	48.5	52.0	67.5
PrepareCoffee	13.0	17.5	15.0	22.0	17.5
RinseSinkBasin	53.5	54.5	62.0	59.0	59.0
ScrubCuttingBoard	24.5	47.0	19.5	20.0	28.5
SearingMeat	10.5	3.5	8.5	11.5	23.0
SetUpCuttingStation	21.5	14.0	18.0	28.0	31.0
StackBowlsCabinet	57.5	69.0	63.5	67.5	69.5
SteamInMicrowave	24.5	12.5	10.0	27.0	29.5
StirVegetables	19.0	5.5	12.5	15.5	24.5
StoreLeftovers	11.5	28.5	28.5	34.5	48.0
WashLettuce	35.5	46.0	53.0	53.0	38.0
Average	22.0	25.6	24.9	29.2	32.8

the future branch reduces the average to 29.2%, a difference of 3.6 percentage points. This advantage of clean-action prediction agrees with the controlled analysis in Section 3, which favors the clean endpoint as a prediction target for high-noise denoising and few-step sampling. Removing future supervision while retaining x -prediction reduces the average to 24.9%, a difference of 7.9 points from the full model. Because future queries are isolated from action tokens and removed at inference (Section 4.4), this comparison supports the value of auxiliary geometric supervision for the shared perceptual representation.

Perceptual conditioning. Final-layer and interleaved conditioning achieve average success rates of 25.6% and 22.0%, respectively, compared with 32.8% for the full model. These results favor sparse feature injection over the two alternative interfaces, consistent with retaining intermediate visual information alongside a common semantic context (Section 4.2). At the task level, final-layer conditioning achieves 47.0% on ScrubCuttingBoard, compared with 28.5% for the full model; x -pred and v -pred each achieve 53.0% on WashLettuce, compared with 38.0% for the full model.

6.3 Real-World Experiments

Platform. We use the mobile manipulation platform described in Section 5.2, comprising a HexFellow Trigger-A3 omnidirectional base and two AgileX PiPER-X 6-DoF arms.

Tasks. The evaluation covers five long-horizon tasks in household, office, workcell, and laboratory scenes. Each task is specified by a single natural-language instruction and requires base repositioning between object interactions. Figure 9 shows representative stages with step-level instructions.

- **Birthday Party Setup.** The robot picks up a cake from the table directly ahead, carries it to the decorated table on the left, and sets it down. It then picks up a birthday candle, passes it between its two hands, and inserts it into the cake. Finally, it moves beside the table and turns on the speaker to play music.
- **Office Folder Arrangement.** The robot picks up a folder from the office desk in front of it, passes it between both hands, and holds it steady. It then carries the folder to the desk on its right, aligns it with the file organizer, and inserts it.



Figure 9. Real-world task execution. Representative stages of the five tasks, from top to bottom: Birthday Party Setup, Office Folder Arrangement, Kitchen Work, Industrial Parts Organization, and Chemistry Lab Operation. Within each task, frames progress from left to right.

- **Kitchen Work.** The robot picks up a food container from the table and pours the food into a pot. It then puts the lid on the storage container, places the container into the cabinet, and closes the cabinet door.
- **Industrial Parts Organization.** The robot places scattered industrial parts neatly into a parts bin, carries the filled bin to the table behind it, and stacks it neatly with the other bins.

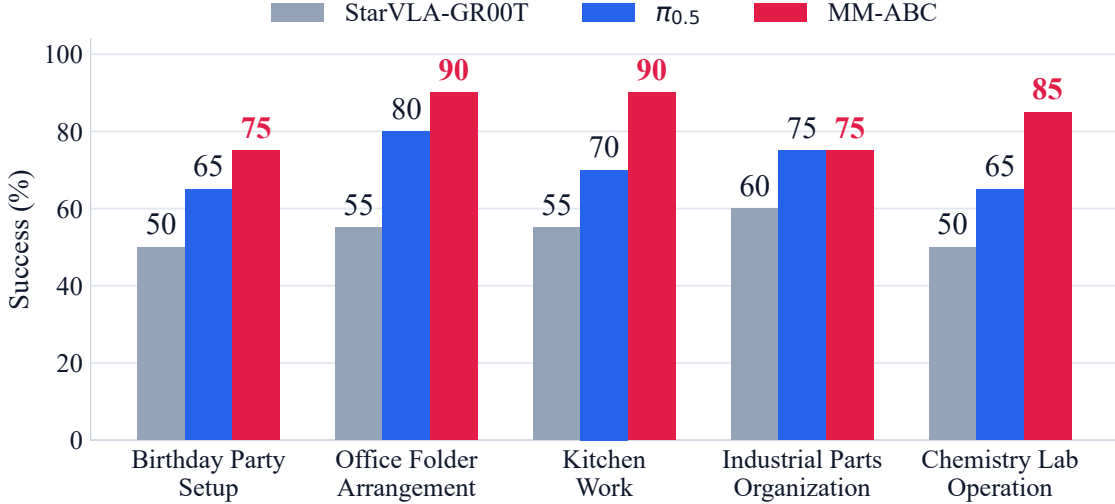


Figure 10. Real-world task success. Success rate (%) over 20 trials per task for each method.

- **Chemistry Lab Operation.** The robot picks up a test tube, moves to the laboratory bench behind it, pours the reagent into a beaker that already holds another reagent, and stirs until the two are thoroughly mixed.

Evaluation protocol. We compare MM-ABC with $\pi_{0.5}$ [20] and StarVLA-GR00T [118], fine-tuning every method on the same 200 teleoperated demonstrations per task. Each method is evaluated over 20 trials per task, and a trial counts as successful only when all steps of the task are completed. We report the percentage of successful trials for each task and the unweighted mean across the five tasks.

Results. As shown in Figure 10, MM-ABC achieves success rates of 75%, 90%, 90%, 75%, and 85% on the five tasks in the order listed above, with a mean of 83%. The corresponding rates are 65%, 80%, 70%, 75%, and 65% for $\pi_{0.5}$ (mean 71%), and 50%, 55%, 55%, 60%, and 50% for StarVLA-GR00T (mean 54%). MM-ABC therefore exceeds the two baselines by 12 and 29 percentage points on average, respectively. The largest margins over $\pi_{0.5}$, 20 points each, occur on Kitchen Work and Chemistry Lab Operation, which combine base repositioning with pouring, lid placement, cabinet-door closing, and stirring. On Industrial Parts Organization, MM-ABC and $\pi_{0.5}$ both succeed in 75% of trials.

7 Conclusion

We presented MM-ABC, a mobile manipulation foundation model built around seeing, coordinating, and imagining arm–base collaboration. Sparse DeepStack conditioning supplies multilevel VLM features to the action expert, and a training-only future branch aligns the shared perceptual representation with future VGGT- Ω geometry without adding inference cost. MM-APT keeps manipulation and body motion in separate streams, couples them through near–far masked joint attention, and predicts clean action chunks. In a controlled synthetic study, clean-action prediction retains less noise in its endpoint estimates, denoises more accurately at high noise, and allocates the arm–body task more accurately under few-step sampling.

Pretrained on more than 5,000 hours from 12 datasets and 17 embodiments, including the self-collected MM-30 dataset, MM-ABC reaches 44.71% success on EBench, a task-weighted 61.2% on RoboCasa365, and the highest mean success on all three ManiSkill-HAB suites. The same architecture reaches 99.1% on LIBERO, and its LIBERO-trained policy transfers to LIBERO-Plus with 82.8% total success. On five real-world tasks, it averages 83% success, compared with 71% for $\pi_{0.5}$ and 54% for StarVLA-GR00T. The from-scratch ablation shows that multilevel conditioning, future supervision, and clean-action prediction each contribute: velocity prediction lowers the composite-seen average by 3.6 points, removing future supervision

lowers it by 7.9 points, and final-layer or interleaved conditioning trails sparse multilevel injection by 7.2 and 10.8 points. These results support treating mobile manipulation as manipulation over a reconfigurable workspace, in which perception, anticipation, and arm–base coordination are learned together.

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